



## CUSTOMER RETENTION ANALYSIS

# Who's leaving, when, and what it costs

Diagnosing repeat-purchase decline for a UK online gift retailer, using two years of real transaction data.

**1.05M**

LINE ITEMS

**5,852**

IDENTIFIED CUSTOMERS

**24 mo**

DEC 2009 – DEC 2011

**40+**

COUNTRIES SOLD TO

## THE SITUATION

# No cancel button, so nobody tells you when a customer leaves

The retailer sells all-occasion giftware to a mix of casual, one-off buyers and wholesale accounts placing large repeat orders. There is no subscription and no cancellation step. A customer simply stops ordering, and the business is often the last to notice.

A drop in repeat purchases could mean the product isn't landing, or it could just mean a strong holiday season pulled in one-time buyers who were never likely to return. Without breaking the data down by who, when, and how behavior changes afterward, those two stories look identical.

## THIS ANALYSIS ANSWERS

- 1 What counts as a lost customer, and how long a silence means churn
- 2 Which customers are gone, drifting, or healthy, specific enough to act on
- 3 Whether newer customers are retained better than earlier ones
- 4 What a churned customer is actually worth

# A vague sense that “retention feels soft” isn’t actionable. A named list is.

1

## Acquisition costs more

Winning a new customer costs more than keeping one, but a retailer that can’t identify who’s at risk has no way to intervene in time.

2

## Churn hides in aggregate

A flat repeat-purchase rate can mask a specific group quietly leaving: a buyer type, a season, a stage in the relationship.

3

## Specific problems are fixable

“Retention feels soft” can’t be campaigned against. A named, sized, dated segment can.

# Two years of transactions, cleaned into a model you can trust

The raw data (online\_retail\_II.csv) is 1,067,371 line items from a UK-based online retailer of all-occasion giftware, covering December 2009 to December 2011 across 40+ countries. Every step from here, cleaning, splitting, scoring, and querying, is logged in assumptions\_log.md so any number in this deck can be traced back to a decision.

01\_data\_cleaning.ipynb stripped non-product codes, cancelled lines, stock write-offs, bad prices, and exact duplicates, taking the raw file down to 1,003,386 clean rows. Orders with no Customer ID (226,794 rows) were set aside as guest checkout, leaving 776,592 rows across 5,852 identified customers for anything that tracks a person over time.

**1.07M**

RAW LINE ITEMS

**1.00M**

ROWS AFTER CLEANING

**5,852**

CUSTOMERS MODELED

## THE PROCESS

1

Clean: pandas removes bad rows, dedupes, and splits guests from identified customers

2

Model: 03\_rfm\_churn\_cohort.ipynb builds RFM scores, the churn cutoff, and cohort labels

3

Query: business\_queries.sql runs the revenue, repeat-rate, and top-spender questions in MySQL

4

Report: validated results feed the Power BI customer dashboard and this summary

WHERE THINGS STAND

# The headline numbers

72.4%

REPEAT PURCHASE RATE

of identified customers ordered more than once

50.7%

CHURN RATE

past 90 days of silence, as of the snapshot date

87%

REVENUE TRACEABLE

to a named customer, the rest is guest checkout

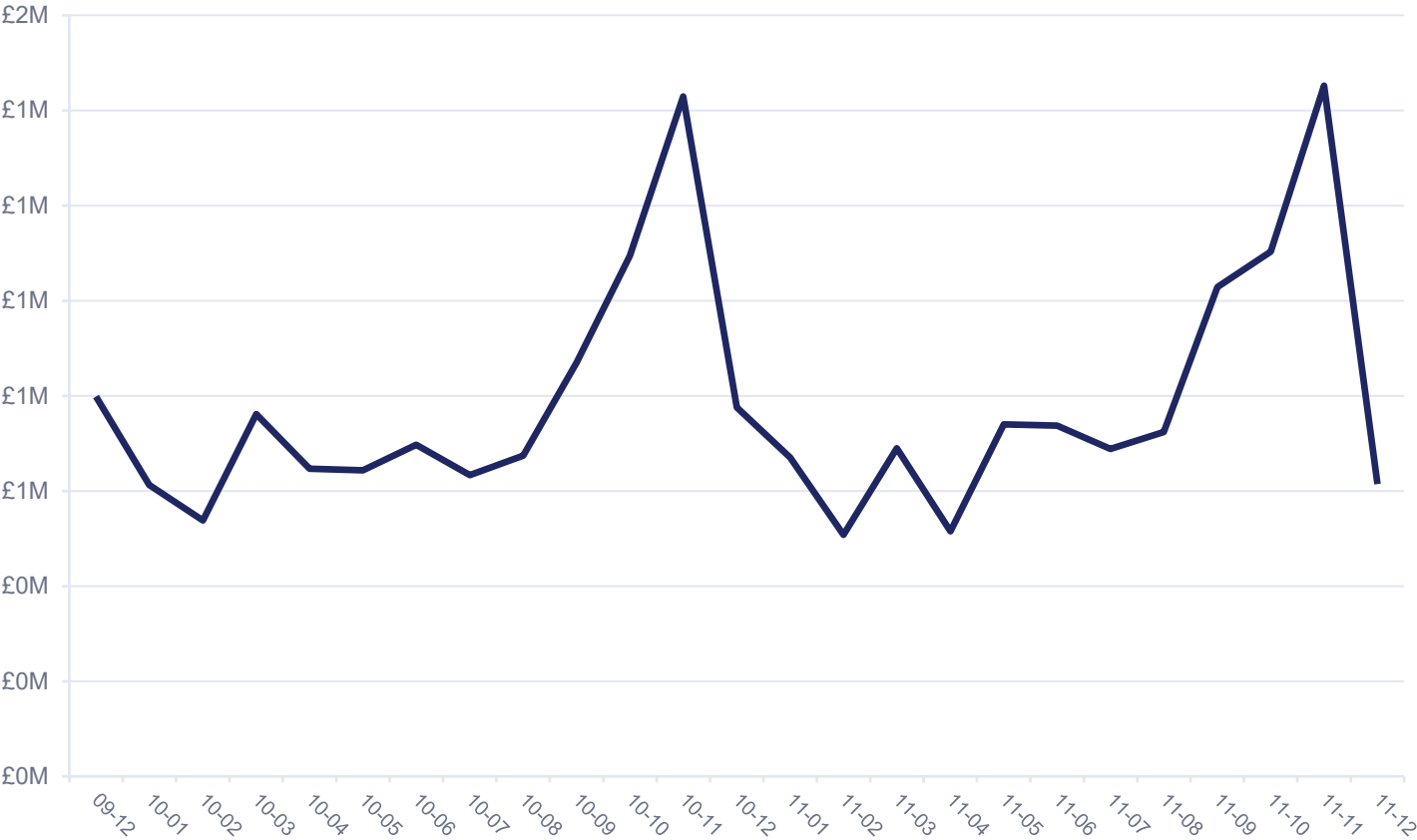
£19.65M

NET REVENUE

across orders\_clean\_full, Dec 2009 – Dec 2011

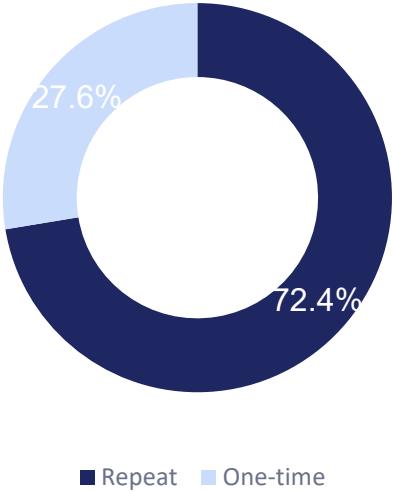
THE BASELINE

# Seasonal, not shrinking: revenue peaks every November



Dec 2011 is partial. The dataset ends Dec 9, so this isn't a real Q4 drop.

REPEAT VS. ONE-TIME

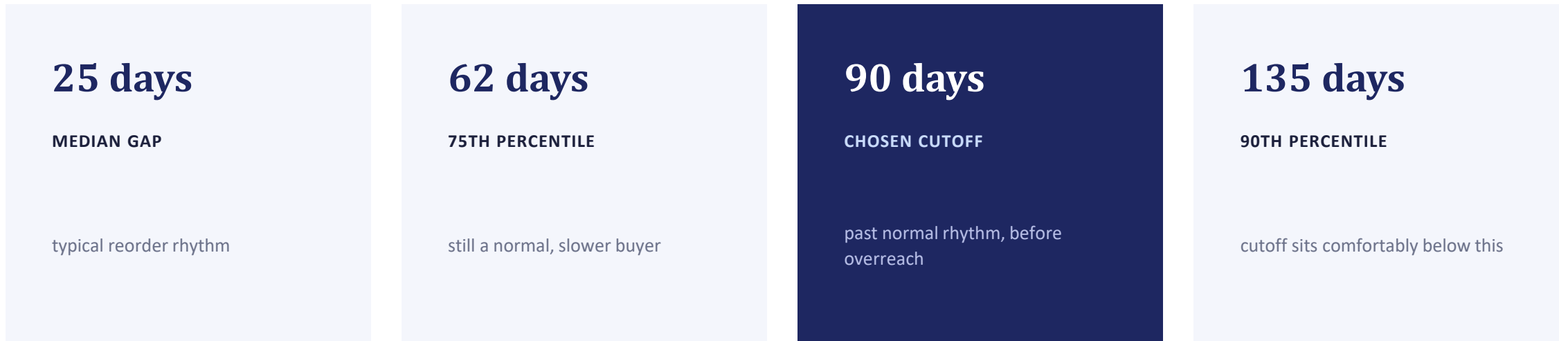


Repeat customers already make up nearly three-quarters of the identified base. The open question is which of them keep buying.

## DEFINING CHURN

# 90 days of silence is the line between “slow” and “gone”

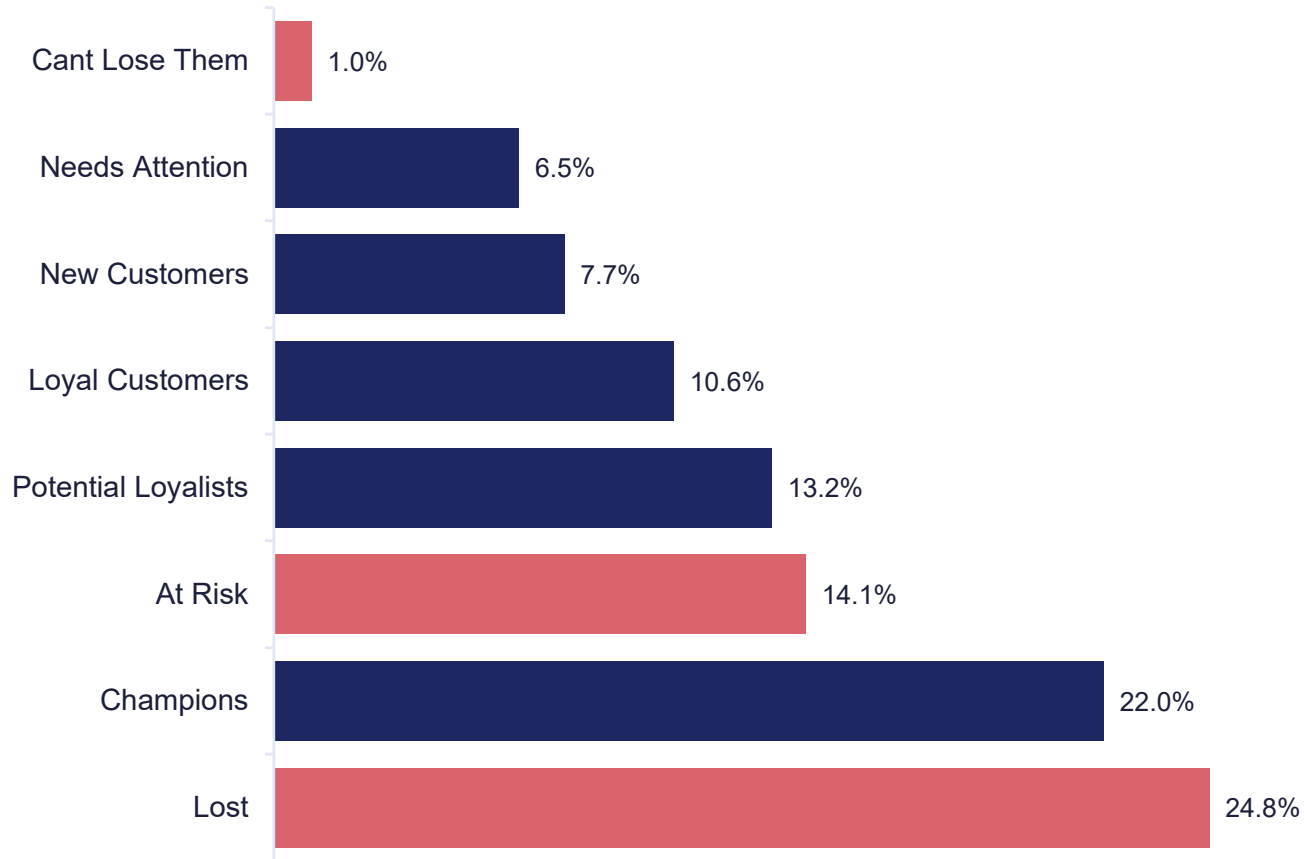
The cutoff wasn't guessed. It comes from the actual gap between consecutive orders across every repeat customer in the dataset.



**50.7%** of the 5,852 identified customers (2,967) are past this cutoff as of the snapshot date. *The most recent 2–3 months are censored: there hasn't been enough time for those customers to lapse, so their churn reads artificially low and shouldn't be compared directly to earlier months.*

## WHO'S CHURNING

# Eight segments, from Champions to Lost



## WORTH FLAGGING

**Lost (24.8%) and Champions (22.0%) are nearly the same size** so the base is genuinely split between customers gone quiet and the retailer's best buyers.

**At Risk + Cant Lose Them = 15.1%** of customers, the group worth acting on first: still active enough to reach, valuable enough to matter.



PRIORITY LIST

# 59 accounts worth protecting first

£3.30M

Value at risk across churned customers

19.35%

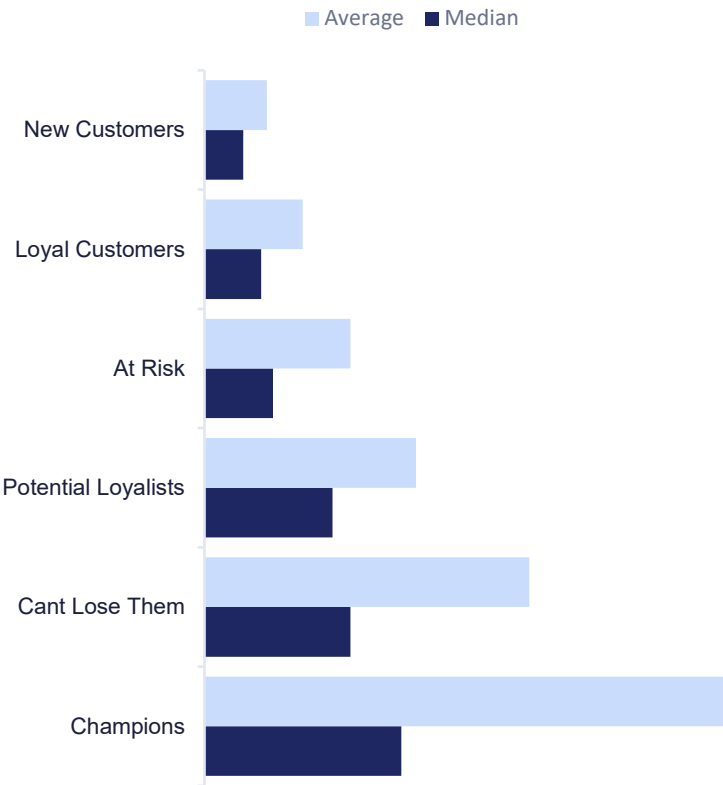
of total customer value already at risk

59

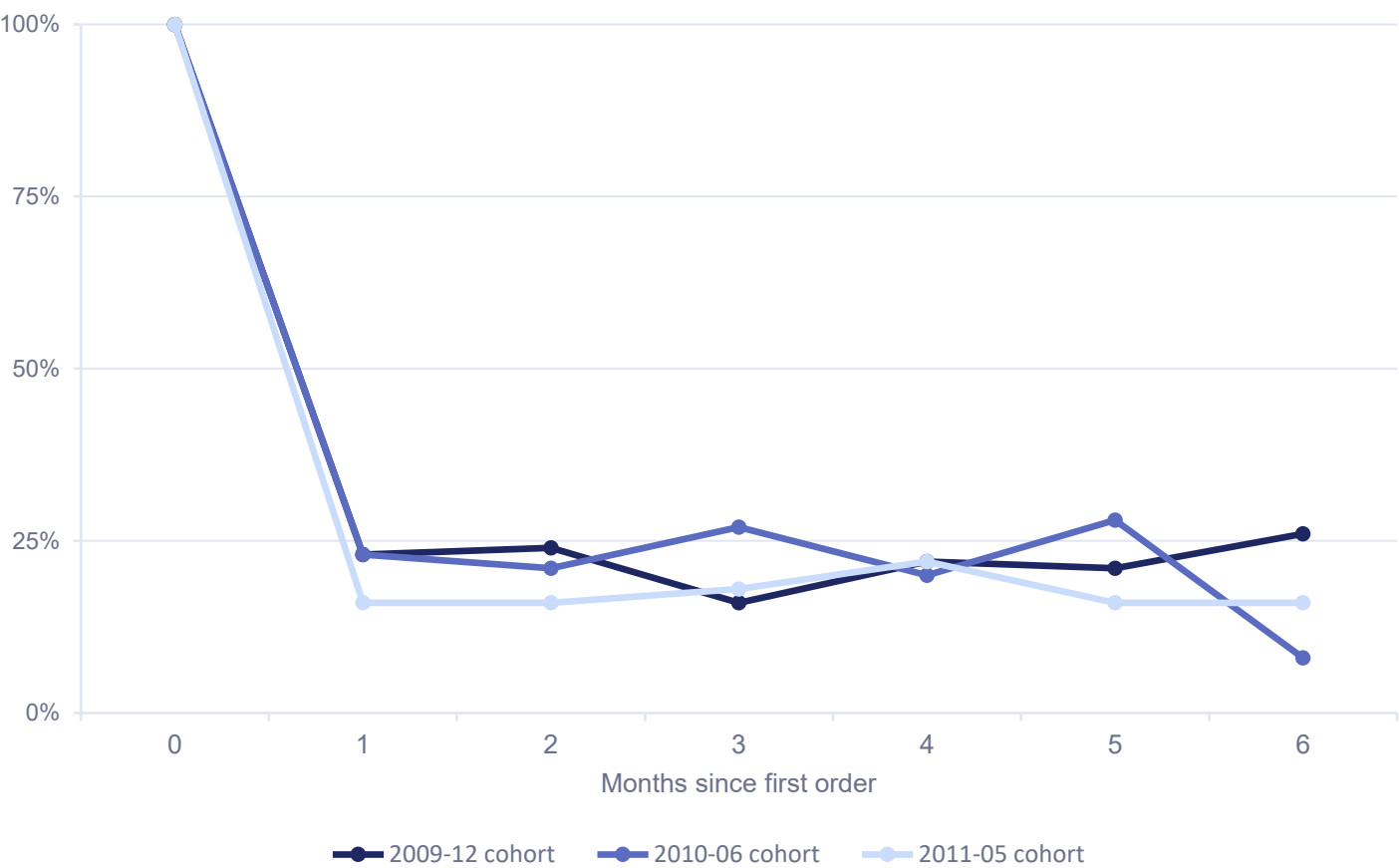
“Cant Lose Them” customers: high value, going quiet

**Two accounts skew every “top customer” view:** Customers 18102 and 14646 each placed 145 orders. That’s a reorder pattern, not casual gift buying, so read them as wholesale, not loyal retail customers.

TYPICAL VS. AVERAGE VALUE



# Most of the drop-off happens in the first two months



READING THIS

**Every cohort loses 70–80% of its customers by month 2**, then the curve flattens, and whoever's left tends to stick around. Retention is largely won or lost in the first eight weeks.

**Scope:** covers *identified customers only* (about 87% of revenue). The last 2 to 3 cohort months are censored: too little time has passed to know if they'll reorder, so don't compare them to older cohorts.

## These numbers describe 87% of the business, not all of it



About a fifth of all orders (22.0% of rows) have no customer identifier attached, so they can't be tied to a person over time. Those orders are kept for revenue and product-level questions, but excluded from anything that tracks an individual, which is also why they carry only 13.1% of revenue instead of 22% of it.

**This isn't just an analysis limitation. It points to a real gap at checkout.** One in five orders isn't being tied to a customer record, which means the business is already flying blind on a fifth of its transactions before any churn question is even asked. Closing that capture gap would make every retention number in this deck more complete going forward.

# Three moves, in priority order

01

## Act on the priority list now

59 “Cant Lose Them” accounts (£3.30M at risk) are still reachable. A targeted win-back campaign has a named list to work from today.

02

## Fix the first 60 days

70–80% of cohort drop-off happens in months 0–2. Onboarding and early follow-up matter more than any later-stage save campaign.

03

## Close the checkout gap

22% of orders carry no customer ID. Capturing identity at checkout turns a fifth of the business from invisible to trackable.

*Full methodology, assumptions, and query results in the accompanying project repository.*



## CUSTOMER RETENTION ANALYSIS

# Thank you

Full methodology, notebooks, and SQL queries live in the project repository. Questions and pushback are welcome.

**59**

ACCOUNTS TO ACT ON

**£3.30M**

VALUE AT RISK

**87%**

REVENUE TRACEABLE

**3**

PRIORITY MOVES